Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18CS32]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER–2021

DATA STRUCTURES AND ALGORITHMS

Time: 3 Hours **Maximum Marks: 70**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.

1. a. **i.** Design a recursive algorithm for computing 2^n for any nonnegative integer n that is based on the formula: $2^n = 2^{n-1} + 2^{n-1}$.

ii. Set up a recurrence relation for the number of additions made by the algorithm and solve it.

iii. Draw a tree of recursive calls for this algorithm and count the number of calls made by the algorithm. **08**

iv. Is it a good algorithm for solving this problem?

b. Define the following asymptotic notations

i. Big Oh

ii. Big Omega **06**

iii. Theta

2. a. Design an algorithm using stack to convert a given infix expression (without parenthesis) into a postfix expression. Give details of any supplementary functions used. **08**

Using the same convert the following infix to postfix. Show each step of the conversion: **A+B/C*D-E^F^H**

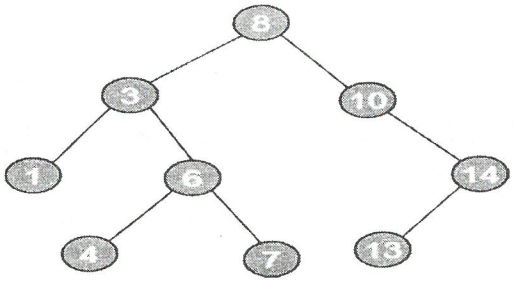
b. Identify the limitation/s of array implemented simple queue. Discuss the possible ways to overcome this limitation? Write an efficient pseudo code for the Circular queue primitive operations. **06**

3. a. Define a Binary Search Tree? Construct the Binary Search tree from the given traversal **08**

Preorder : **A B D G H C E I F**

Inorder : **G D H B A E I C F**

b. Discuss the deletion operation (routine/function) on a binary search tree. Perform the same to delete 3 in the given binary search tree. **06**



OR

4. a. Define Hashing? Show the resulting hash table using

   i. Quadratic probing

   ii. Double hashing (second hash function h2(x) = (x-2)mod 5 **08**

   On the following input {65, 52, 48, 17, 25 } and hash function h(x) = x mod 8

   b. Write an iterative function to perform the preorder traversal. **06**

5. a. Design an algorithm used to perform partition used in Quick Sort. Apply the same to sort the following elements **08**

   8 4 2 10 7 12 1 15

   b. Write an algorithm or pseudocode for radix sort. Apply the same on the following elements **06**

   329, 457, 657, 839, 436, 720, 355, 124.

OR

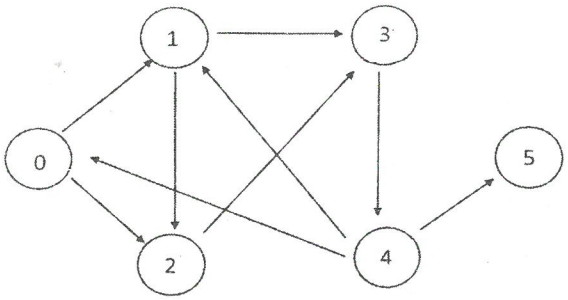
6. a. Define a heap. Design an algorithm to construct a max heap. Construct a max-heap for the given set of elements using bottom-up approach **08**

   5 8 7 6 10 4 1

   b. Design an algorithm to merge two sorted arrays. Analyze the efficiency of the same. **06**

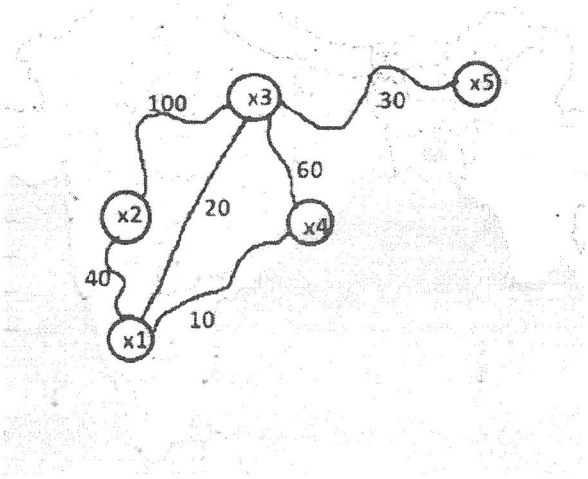
05

7. a. i. Discuss the DFS traversal algorithm. Perform the same on the given graph



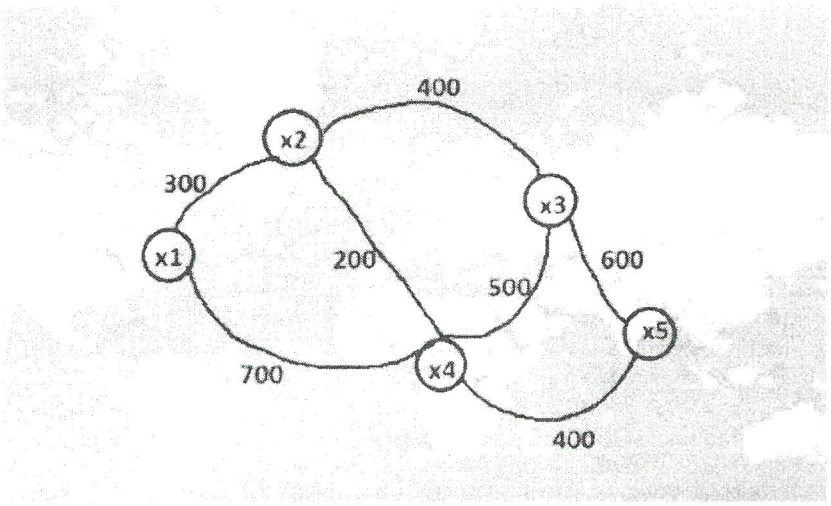
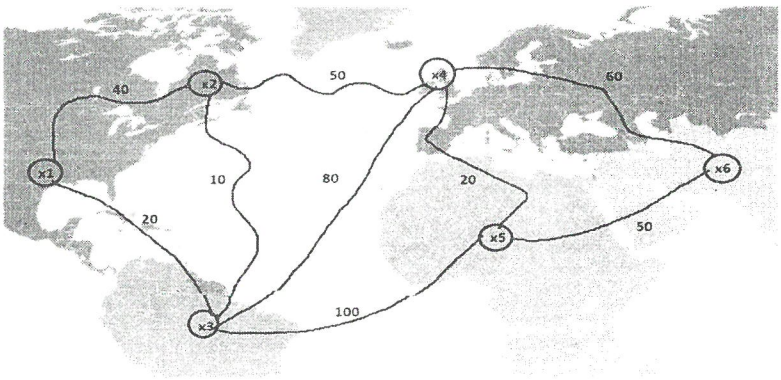
   ii. Differentiate between DFS and BFS traversal. **03**

   b. Write the Prim's algorithm and give its efficiency. Apply Prims's algorithm to find minimum cost spanning tree from vertex "x1". **06**



OR

8. **a.** Write the Dijkstra's algorithm and give its efficiency. Apply Dijkstra's algorithm to find single source shortest paths from source vertex "x1". **08**



b. Define minimum spanning tree. Apply kruskal's algorithm to find the same for the graph given below. **06**